

Elements of Position

The Geometric Kernel

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What the walk measures

Paper I named the closed unit and the pair. This paper is the walk that produces those measurements from two operators and a count. Nothing is loaded into the operators. What is named after the fact — a half-turn length, a leftover of two readings, a ratio of chord to path — is read off the tables the walk writes.

The source is `Elements_of_Position.Geometric_Kernel.py`. The tables are `eop_crossings.csv`, `eop_pairs.csv`, `eop_summary.csv`, from a run with shear $h = 0.002$ and 400 horizon hits. Those numbers are a sheet size and a depth of count. The identities do not belong to that sheet.

A note. Claims that can be proved from the operators are proved as identities. Claims that are counts are reported from the tables. The two are kept apart.

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The two operators

Start at $(1, 0)$. A heading is a pair of coordinates whose length after the bind is 1.

Definition 1 (Shear). For a step size $h > 0$,

$$T_h(x, y) = (x - hy, y + hx).$$

Definition 2 (Bind).

$$N(x, y) = \frac{1}{\sqrt{x^2 + y^2}}(x, y).$$

Definition 3 (Step).

$$\text{step}(x, y) = N(T_h(x, y)).$$

If the last state already had length 1, the un-bound length after the shear is an identity of the sheet:

$$|T_h(x, y)| = \sqrt{1 + h^2}.$$

The bind leftover of one kick is therefore

$$\sqrt{1 + h^2} - 1,$$

constant in the step index. On the recorded run this is 1.999998×10^{-6} . The Euclidean distance between two successive bound headings is likewise a function of h only. On the recorded run it is 0.001999997. Those two numbers describe the grain of the walk. They are not a position on the pair.

The horizon count

A hit is the heading crossing the line $y = 0$. Down: $y > 0$ then $y \leq 0$. Up: $y < 0$ then $y \geq 0$. Each hit is one half-turn of the heading.

Definition 4 (First gap). H_0 is the step index of the first hit.

On the recorded run $H_0 = 1571$, side down, heading

$$(-0.99999992, -0.00040316).$$

The heading is on the left horizon to one kick of vertical.

Definition 5 (Snap scale). The m -th hit has snap scale $k = m$.

The frozen scale $k_{\text{frozen}} = j/H_0$ uses the first gap as a fixed ruler. The two scales agree at $m = 1$. They part when later gaps are not exactly H_0 .

Gaps

After the first hit the tables show only two gap values, 1570 and 1571. Histogram on 400 hits:

$$1571 \text{ occurs } 320 \text{ times,} \quad 1570 \text{ occurs } 80 \text{ times.}$$

Mean gap:

$$1570.8.$$

The length a continuous turn of this shear would take is $\pi/\text{atan}(h) = 1570.798421$. The discrete walk cannot land on that length, so it pays with the two neighboring integers. The mean tracks the continuous half-turn to three decimal places.

Observation 1. $H_0 \cdot h = 3.142$ on this sheet. That product is the half-turn read as steps times step size. It is not inserted into T_h or N .

Heading at the hits

At every recorded hit the heading sits on a horizon. Odd m (down): $h_x \approx -1$, h_y in $(-0.002, 0]$. Even m (up): $h_x \approx +1$, h_y in $[0, 0.002)$. The vertical coordinate never grows with m . Sampling at the cross keeps the heading on the wall. Sampling at $m \cdot H_0$ does not: that ruler stacks the first-crossing leftover and the heading walks off the axis.

The pair

At snap scale $k = m$,

$$r_{\text{out}} = \frac{k}{2}, \quad r_{\text{in}} = \frac{2}{k}.$$

Proposition 1. *For every $k > 0$,*

$$r_{\text{out}} r_{\text{in}} = 1.$$

Proof. Immediate. □

Proposition 2.

$$\text{GM}(r_{\text{out}}, r_{\text{in}}) = 1.$$

Proof. $\sqrt{r_{\text{out}} r_{\text{in}}} = \sqrt{1} = 1$. □

The crossings table records $\text{gm} = 1$ on every row, to machine precision. One row at $m = 374$ prints 0.9999999999999999. That is the same identity written in floating point.

Definition 6 (Leftover of the pair).

$$\tau(k) = \text{AM} - \text{GM} = \frac{k/2 + 2/k}{2} - 1 = \frac{k}{4} + \frac{1}{k} - 1.$$

Proposition 3. $\tau(k) = 0$ if and only if $k = 2$ (for $k > 0$).

Proof. $\frac{k}{4} + \frac{1}{k} = 1$ multiplies to $k^2 - 4k + 4 = 0$, hence $(k - 2)^2 = 0$. □

The table: $m = 2$, $\tau = 0$, $r_{\text{out}} = r_{\text{in}} = 1$, side up, $j = 3142 = 2H_0$. That is the kiss. At $m = 1$, $\tau = 1/4$. At $m = 4$, $\tau = 1/4$ again.

Proposition 4. $\tau(k) = \tau(4/k)$ for every $k > 0$.

Proof. Substitute. □

The walls swap:

$$r_{\text{out}}(k) = r_{\text{in}}(4/k), \quad r_{\text{in}}(k) = r_{\text{out}}(4/k).$$

The pairs table records this at $m = 1$ with partner $m = 4$: leftover 0.25 on both sides, walls (0.5, 2) against (2, 0.5). At $m = 2$ the state is its own partner.

The pairs file as written only keeps partners that round to an integer snap inside the run. For $k > 4$ the partner $4/k$ lies below 1 and is discarded by that rounding. The identity $\tau(k) = \tau(4/k)$ does not need the partner to be an integer snap. It is an identity of the pair formula.

Chord and path of one half-turn

Place the outward wall on the heading:

$$P_m = r_{\text{out}}(m) (h_x, h_y).$$

The straight line from P_{m-1} to P_m is the leg chord. The summed micro-path between those hits is the leg path. The table column `leg_chord_over_path` is their ratio.

Proposition 5. *If the heading at hit $m - 1$ is the opposite of the heading at hit m , and the radii are $(m - 1)/2$ and $m/2$, the chord is*

$$\frac{m-1}{2} + \frac{m}{2} = m - \frac{1}{2}.$$

Proof. Opposite unit headings u and $-u$:

$$|(m/2)(-u) - ((m-1)/2)u| = (m/2 + (m-1)/2)|u| = m - \frac{1}{2}.$$

□

The table: $m = 2$ chord 1.500000; $m = 3$ chord 2.500000; $m = 4$ chord 3.500000. The pattern holds. The ratio of that chord to the summed path climbs toward $2/\pi$. Selected rows:

m	chord/path	$2/\pi$
2	0.62243	0.63662
4	0.63402	
10	0.63665	
20	0.63689	
100	0.63695	
400	0.63654	

The half-turn of a circle has diameter over semicircle equal to $2/\pi$. That is what the long baseline is reading. The small oscillation at large m is the leftover angle of a discrete hit, not a second constant.

The leftover integrated

From the kiss outward the leftover of the pair is a region between the two walls and the axle $GM = 1$. Its integral in the snap scale is

$$\int_2^N \tau(k) dk = \left[\frac{k^2}{8} + \ln k - k \right]_2^N = \frac{N^2}{8} + \ln N - N + \frac{3}{2} - \ln 2.$$

On the recorded run $N = 400$:

$$\frac{400^2}{8} + \ln 400 - 400 + \frac{3}{2} - \ln 2 = 19606.798317.$$

The summary table:

$$\begin{aligned} \text{wedge_from_crossings} &= 19606.798317366545, \\ \text{wedge_closed} &= 19606.79831736655, \end{aligned}$$

difference -3.6×10^{-12} . The script sums the same antiderivative from hit to hit. The match is the identity, not a numerical accident.

When the leftover is instead accumulated as a rectangle sum along a frozen- H_0 walk, the difference from the closed form at large N is

$$\frac{\text{discrete} - \text{closed}}{N h} \approx \frac{1}{8\pi}.$$

Check at an earlier frozen run, $N = 3600$, $h = 0.002$:

$$\frac{0.28612}{3600 \times 0.002} = 0.03974, \quad \frac{1}{8\pi} = 0.03979.$$

That ratio is the staircase error on the $k^2/8$ term once $H_0 \sim \pi/h$. It is a property of summing on the frozen ruler. The snap integral does not carry it.

What the three files hold

eop_summary.csv

key	value
h	0.002
H_0	1571
half-turns	400
sheet leftover	1.999998×10^{-6}
heading chord	0.001999997
$\pi/\text{atan}(h)$	1570.798421
$H_0 \cdot h$	3.142
wedge (snap)	19606.798317
wedge (closed)	19606.798317
mean gap	1570.8
gap min / max	1570 / 1571
histogram	{1571 : 320, 1570 : 80}

Sheet: h , leftover of one kick, heading chord, H_0 . Object: product of the walls, leftover formula, kiss at 2, mean gap tracking the continuous half-turn, wedge identity.

eop_crossings.csv

Four hundred rows, one per hit. Forced on every row: $g_m = 1$, $r_{\text{out}}r_{\text{in}} = 1$, $\tau = k/4 + 1/k - 1$, heading on a horizon. Side alternates down, up, down, up. Frozen scale at $m = 400$ is 399.949, snap scale is 400. The ruler j/H_0 has slipped by five hundredths of a half-turn. The heading has not.

eop_pairs.csv

Integer partners of $k \leftrightarrow 4/k$ that land inside the run. The only clean integer pair in $1 \leq k \leq 400$ is $1 \leftrightarrow 4$, and the self-pair $2 \leftrightarrow 2$. Rows that list partner $m = 1$ for $k > 4$ are the rounding rule, not the involution. Read the involution from the formula and from $m = 1, 2, 4$.

What stands

The operators produce a heading that counts its own half-turns. The count is two neighboring integers whose mean is the continuous half-turn of that shear. Scale is the hit index. The pair at that scale has product 1 and leftover $\frac{k}{4} + \frac{1}{k} - 1$, zero only at 2. Chord over a half-turn of the outward wall, against the path of that same half-turn, reads $2/\pi$. The leftover integrated from the kiss is the elementary antiderivative of τ . A frozen first-crossing ruler rotates the heading off the axis; a snap to the axis does not.

Named after the measure, not before it: $H_0 \cdot h$ as a half-turn length; $2/\pi$ as diameter over semicircle; $1/(8\pi)$ as the frozen-sum error. None of those names enters T_h or N .

What is not granted: a last decimal as closure of the heading; a completed run of the count; a leftover of the pair treated as a capacity or a field.

The first sentence of Paper I is still the sentence. The kernel is how the walk answers it.